

Representation and Framing Bias in CC-News Coverage of Feminist Protests

Chris Xia, Cristina Gomero, GengXin Geng, Greta Pfeiffer, Laura Ponoran

University of Amsterdam

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Executive Summary

This study examines how major feminist protest movements are represented in mainstream news media via a mixed-method approach consisting of quantitative text-analysis methods and qualitative interviews. The limitations of these methods are audited throughout this paper and discussed within the greater realm of algorithmic bias. Drawing on the protest paradigm, framing theory, and feminist critical theory, the research investigates whether feminist protests are systematically marginalized in news coverage and how media attention influences the visibility of feminist issues, particularly through mechanisms that may reinforce existing power structures and the socio-political status quo.

Two research questions guided the study. First, how are feminist protests portrayed across different news outlets, regions, and events? Second, how can the distribution and sentiment of news coverage be used to understand broader patterns of feminist inequality? The study hypothesized that feminist protests would be underrepresented and framed negatively (H1), and that mainstream media would dilute protest visibility by embedding it within dominant news agendas that reinforce the status quo (H2).

A mixed-methods approach was used. Approximately 7,500 articles covering ten major feminist protest events between 2017 and 2023 were collected from the CC-News archive and analyzed using BERTopic topic modelling and VADER sentiment analysis. Quantitative findings were supplemented with desk research and seven semi-structured interviews.

The results provide strong support for H2. Feminist topics received substantially more attention during protest periods (12.9% of topic distribution) than during control weeks (1.6%), but remained overshadowed by dominant news categories such as geopolitics, institutional politics, and sports. This suggests that feminist movements are not excluded from media coverage but are structurally marginalized within the broader news environment.

Support for H1 was mixed. Sentiment varied considerably across events, languages, and publishers, with some protests receiving neutral or positive coverage. However, large differences between language groups revealed limitations in the sentiment analysis model, highlighting the importance of auditing computational methods alongside media content.

Overall, the study finds that feminist protests receive increased media attention during periods of mobilization but remain overshadowed by dominant mainstream news agendas, supporting arguments about the reinforcement of the status quo. The results also demonstrate that computational text-analysis methods can introduce important biases, underscoring the need for transparency and auditing when using such tools to analyse social issues.

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Introduction and Problem Statement

In modern liberal democracies, a persistent divergence remains between legal guarantees of equality and the lived experiences of marginalised populations. When state institutions fail to adequately dismantle deeply rooted patriarchal structures, feminist movements are often driven to shift their strategies from institutional involvement to mass public mobilisation (Portillo & Bono, 2021), which are susceptible to how the press chooses to portray them.

Analysing these media dynamics is increasingly important as computational social science shifts toward large-scale, automated text analysis. However, algorithmic transparency alone remains insufficient for guaranteeing accountability because the complexities of these frameworks are difficult for the public and policymakers to fully comprehend (Ananny & Crawford, 2018). Machine learning models are not neutral technical tools; they are designed by researchers and trained on data that reflect existing societal hierarchies (Shah et al., 2020; Santy et al., 2023).

To address these challenges, this study implements a computational pipeline combining BERTopic topic modelling and VADER sentiment analysis to map the thematic distribution and emotional framing of news coverage across ten large-scale international feminist protest events within the CC-News archive. Rather than focusing on algorithmic neutrality as an abstract ideal, the study uses these tools to examine how mainstream news content represents global feminist struggles and how computational methods shape that analysis (Poell, 2020).

Research Question 1 (RQ1)

How are major feminist protest movements portrayed in the news, and are there measurable differences in language across outlets, regions, and events?

Related Hypothesis (H1)

Mainstream news media coverage frames feminist protests through delegitimizing linguistic patterns, resulting in highly negative sentiment connotations and structural underrepresentation. This aggregate bias is further compounded by model-level errors: lexicon-based sentiment tools introduce systemic artefacts when analysing non-English text, and unsupervised density-based clustering models systematically relegate localised protest themes to unclassifiable outlier documents.

Research Question 2 (RQ2)

How can the distribution and connotation of news content be used to reflect on feminist inequality from the perspective of protests critically?

Related Hypothesis (H2)

Media institutions minimize the disruptive potential of public mobilization by embedding feminist protest coverage within a high volume of normalized, non-protest news categories such as sports, business, and institutional politics. This thematic distribution provides empirical validation for status quo bias, illustrating how dominant media networks dilute radical systemic demands to preserve existing socio-political arrangements.

Background and Policy relevance

Algorithmic Bias and Social Inequality

Media representations play a crucial role in shaping public understanding of social movements and political demands (Gasparyan et al., 2025). Feminist protests are particularly vulnerable to biased forms of representation because they challenge dominant social norms and existing power structures. The protest paradigm suggests that mainstream news media often portray protests through frames emphasizing conflict, disruption, or violence rather than activists' political demands. As a result, feminist movements may be delegitimised, marginalised, or reduced to simplified narratives obscuring their broader social and political goals. For a more in-depth discussion of the status quo, see Appendix A.

Policy Implications and Structural Inequality

The governance of information systems has become an important policy issue as governments, public institutions, and technology companies increasingly rely on automated systems to understand public discourse. Concerns regarding transparency, accountability, fairness, and discrimination have motivated regulatory initiatives such as the European Union's AI Act and GDPR (see Appendix I) (Aysolmaz et al., 2023). This study contributes to these discussions by showing that biases relevant to media representation and social inequality can emerge throughout the analytical pipeline, including data collection, source selection, preprocessing decisions, and model design. Language bias, event selection bias, and topic visibility bias all influenced the final outputs of the analysis, reinforcing the importance of treating computational results as socially and methodologically constructed rather than neutral representations of reality (Metzler et al., 2024).

Beyond technical governance, these findings are also relevant for public institutions and policymakers because media visibility directly shapes which social issues are perceived as politically salient. Governments rely on media ecosystems as indirect signals of public attention and societal priorities. In this study, feminist protest coverage is shown to increase substantially during mobilization periods, while remaining limited in everyday news environments and consistently overshadowed by dominant topics such as geopolitics, institutional politics, and

sports. This uneven distribution matters for democratic representation, as it affects the sustained visibility of feminist issues in public discourse beyond moments of protest.

These patterns have uneven consequences across stakeholder groups, particularly feminist movements, journalists, media institutions, policymakers, and news consumers. Feminist movements depend on media visibility to communicate, mobilize support, and influence public debate (Carvalho, 2025). When media systems prioritize dominant narratives, feminist actors may experience reduced visibility within broader information environments.

From a feminist perspective, these visibility patterns reflect broader inequalities in gender, power, and representation embedded within institutional and cultural structures rather than individual attitudes (Cislaghi, 2020). The topic modeling results show that mainstream news agendas consistently dominate over feminist protest narratives, while language-related disparities suggest that computational systems may further privilege dominant linguistic groups. Together, these findings highlight how both media structures and analytical tools shape which social movements become visible in public discourse, directly linking to the research questions on representation and inequality in news coverage.

Theoretical Framework

The project is grounded in social science theories that explain how media representations shape public understanding of social movements.

Framing Theory argues that media organizations selectively emphasize particular aspects of events while downplaying others. This process influences how audiences interpret political and social issues. In the context of feminist protests, framing theory helps explain variations in how different media outlets portray protest goals, legitimacy, conflict, or violence (Arowolo, 2017; Crenshaw, in Wang et al., 2024).

The Protest Paradigm further suggests that media coverage often emphasizes disruption and confrontation rather than protesters' political demands. This framework is particularly relevant for understanding why feminist protests may receive negative or delegitimizing coverage across news organizations. (Harlow et al., 2023)

While traditional theories explain patterns of media representation, critical theory provides tools for examining the power structures that produce these patterns. Feminist theory is particularly important because it views gender inequality as embedded within institutions, cultural norms, and systems of knowledge production rather than merely individual attitudes (Crenshaw, in Wang et al., 2024; Haraway, in Hancox-Li & Kumar, 2021).

From a feminist perspective, algorithmic bias is a reflection of broader social power relations. Computational systems learn from historical data that already contains inequalities and dominant

narratives. Consequently, algorithms may reproduce existing forms of marginalization while appearing objective and neutral.

This perspective is reinforced by Noble's concept of algorithmic oppression, which demonstrates how digital systems can reinforce social inequalities, and by Couldry and Mejias' theory of data colonialism, which argues that contemporary data infrastructures often privilege dominant cultural and linguistic norms while marginalizing alternative forms of knowledge (Noble, 2018; Couldry & Mejias, 2019). These ideas are directly reflected in our findings regarding language bias and the invisibility of feminist topics within broader news discourse.

By combining traditional social science theories with feminist critical perspectives, the project moves beyond identifying patterns of representation and instead examines how computational systems participate in the reproduction of power, inequality, and visibility within digital media environments.

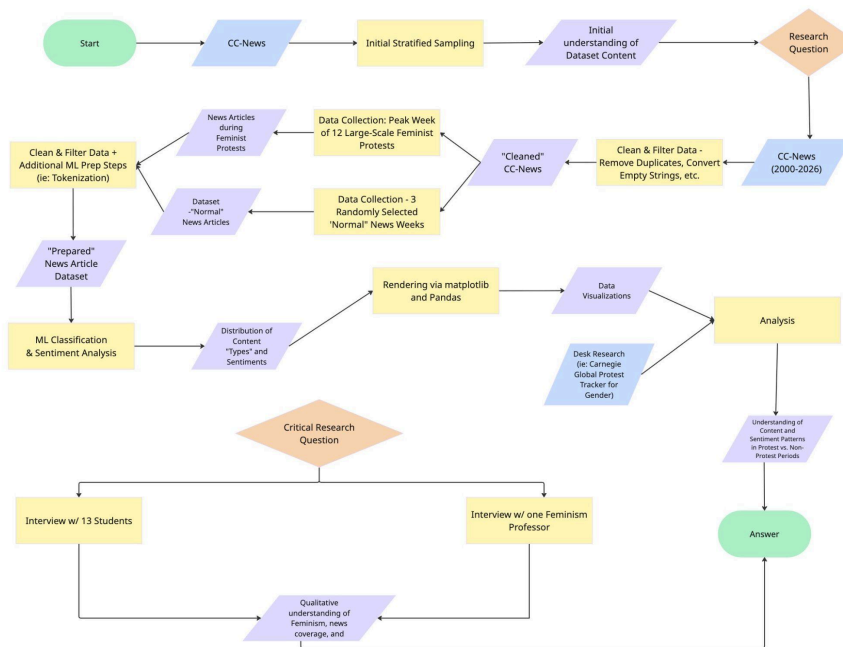
Data and Methods

Research Design

This study implements a mixed-methods approach not only to statistically validate its findings, but also to develop a more nuanced understanding of the general atmosphere surrounding feminist protests and their subsequent news coverage. More specifically, the quantitative aspect determines patterns in topic distribution and sentiment across feminist protest and non-protest periods, while the qualitative element- consisting of desk research and interviews- contextualizes these patterns within feminist theory and real-world experience. The complete pipeline, from data collection to final analysis, is outlined in Figure 1.

Figure 1

Data Science Flow Diagram



Dataset

The quantitative aspect draws on data from CC-News, an open-source repository of international news articles upkept by the Common Crawl Foundation (*Common Crawl - Open Repository of Web Crawl Data*, 2026). CC-News is widely used in NLP research due to its broad scope and language diversity. Working directly with raw CC-news data allows for the elimination of intermediary bias introduced by third-party data providers; that is, any analytical decisions are our own and are made transparent throughout this section. This is summarized in Appendix B.

Articles were collected using the Fundus library and its CCNewsCrawler, which streams articles directly from the CC-News archive. Fundus was chosen over alternative collection methods because it relies on manually crafted, publisher-specific parsers that produce clean text extraction and returns structured fields- including title, body text, authors, topics, and publication date- with minimal additional processing required (Dallabett, et al., 2024).

Data Collection

Event Selection

Articles were collected from the peak coverage week of ten large-scale international feminist protest events from 2017 to 2023. For a full list of protest events and timelines, see Table 1. Events were selected from the Carnegie Global Protest Tracker, a continuously updated database documenting significant anti-government protest movements worldwide since 2017, and were prioritized based on their scale, international relevance and visibility, and the diversity of their surrounding political contexts (Carnegie Endowment for International Peace, 2026).

The “peak week” for each event was selected based on the event type. For single-day or recurring events, the event date itself was positioned on day two or three of the seven-day window so as to capture pre-event buildup, as the event was anticipated, and the subsequent wave of sporadic media coverage. This approach follows event-study principles, which recommend constructing observation windows that account for both pre-event information effects and post-event reactions (MacKinlay, 1997). For sustained or ongoing protests, the window began on the day of the kick-off event in order to capture the initial influx of media attention and the early development of the news cycle surrounding the protest.

Table 1. Article Collection Time-Frames

Protest Event	Collection Time Frame
Women’s March	January 19 - 26, 2017
International Women’s Strike	March 6 - 13, 2017
#MeToo Protest	October 15 - 22, 2017
International Women’s Day (IWD)	March 6 - 13, 2018
Swiss Women’s Strike	June 12 - 18, 2019
Polish Women’s Strike	October 22 - 28, 2020
Sarah Everard Vigils	March 11 - 18, 2021
Roe v. Wade Leak Protest	May 3 - 10, 2022
Women, Life, Freedom (Iran)	September 16 - 23, 2022
Israeli Women’s Protests	February 27 - March 6, 2023

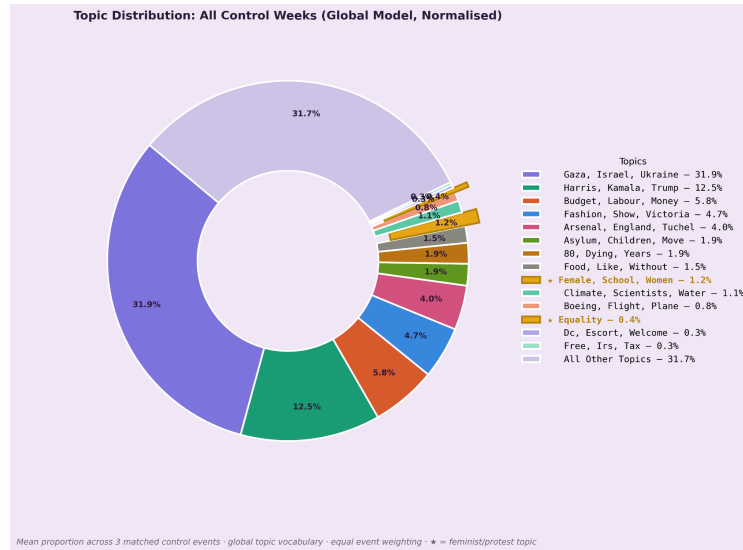
Ideally, each protest event would have been paired with a unique matched-control period to maximize comparability. Matched comparison designs are commonly used to improve the validity of analyses by providing a more closely related, unbiased baseline (Lopez Bernal et al., 2018). However, due to computational constraints, a full matched-control design was not feasible for this study. Therefore, three randomly selected control weeks from non-protest periods were collected. The topic breakdown during these weeks can be seen in Figure 2.

It is worth noting that article counts per event vary substantially (Figure 3). The four pre-2019 events- namely, the Women's March 2017, International Women's Strike 2017, #MeToo 2017, and IWD 2018- contain fewer articles due to sparser CC-News crawl coverage prior to

mid-2019; consequently, the per-publisher collection cap of 50 articles was not reached. The full regional and language breakdown of the dataset is shown in Figure 4, while the language composition of individual publishers is detailed in Figure 5.

Figure 2

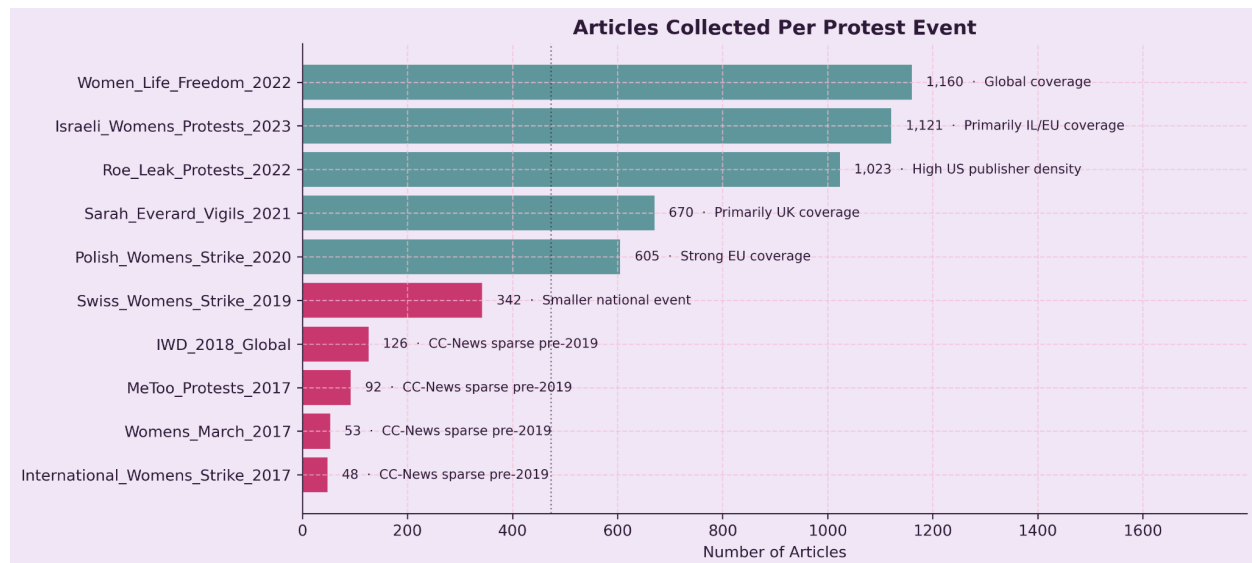
Topic Distribution of Control Weeks - Normalized



Note. “Normalized” refers to scaling the news content from a respective protest week to reflect how many articles were available during that time period so as to not overwhelm or misrepresent the true coverage.

Figure 3

Articles Collected per Protest Event



Quantitative Methods

Data Preprocessing

Raw articles enter the processing pipeline as multilingual text and undergo six preprocessing stages before being used for sentiment analysis or topic modeling.

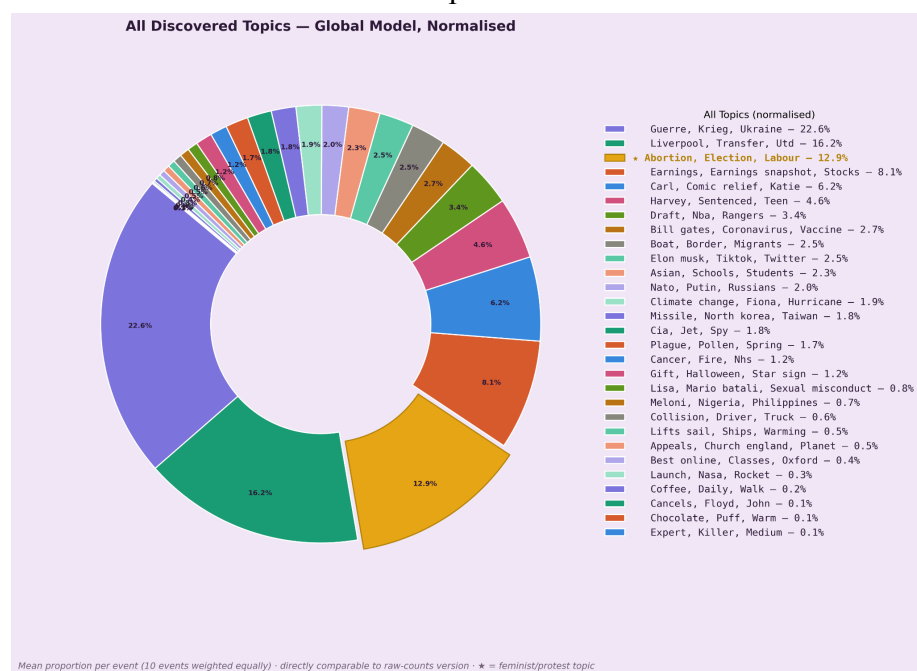
First, each article is passed through Lingua language detector, which has been pre-configured to expect one of the four languages present in the dataset. Non-English articles are machine-translated into English using Helsinki-NLP’s opus-mt models, and the translated text is stored alongside the original. Afterward, downstream steps operate solely on English text. This pre-translation was intentional, as it corrected earlier pipeline failures in which even a multilingual embedding model produced homogenous clusters with large quantities of stop words due to within-language similarity (as seen in Appendix H, panels D-E) (Artetxe & Schwenk, 2019).

Afterwards, the text is passed through a cleaning step to remove residual HTML tags and other formatting discrepancies from titles before they reach BERTopic, in order to prevent contamination of topic labels. Topic clustering is then performed solely on article titles, not body text. Initially, modeling was conducted on body text, however, due to a prevalence of standard vocabulary, topics were overly generic and not meaningful. Consequently, titles were used instead, as they can be considered concise summaries of article content, yielding cleaner results. The difference between these two approaches can be seen in Appendix H, panel A.

Following these preliminary steps, four separate countermeasures were applied to reduce stopword dominance. These include a custom multilingual stopword list, a corpus-frequency ceiling to exclude terms appearing in a large proportion of documents (85% for the global model, 60% per event), a minimum document frequency of two, a minimum token length of three characters, and BERTopic’s ClassTfidfTransformer with BM25, which also down-weights terms frequent across both topics and documents (Leskovec et al., 2014; Manning et al., 2008).

Once topics were assigned, the full dataset contained 7,500 articles, which were modeled with a minimum topic size of three, resulting in hundreds of micro-topics. Therefore, hierarchical reduction was implemented to merge these into 30 macro-themes for the global model. This step is omitted in per-event models to preserve smaller, more focused topics and improve granularity (Blei et al., 2003). Raw BERTopic labels can encode the top three keywords in varying arbitrary orders; therefore, this was normalized via alphabetical sorting and a Maximal Marginal Relevance representation model to penalize highly similar keywords, ensuring final labels are distinct yet representative. Outlier labels are collapsed into a single “All Other Topics” category (Grootendorst, 2022; McInnes et al., 2017). This output can be found below in Figure 6.

Figure 6
Distribution of Clustered News Topics



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Note. “Normalized” refers to scaling the news content from a respective protest week to reflect how many articles were available during that time period so as to not overwhelm or misrepresent the true coverage.

Sentiment Analysis

Sentiment scoring was performed using VADER, a method specifically designed for social and journalistic text, which produces a score ranging from -1 to $+1$, representing maximum negativity and positivity respectively for each article (Hutto & Gilbert, 2015). All scoring was conducted on translated English text to ensure consistency. VADER was chosen due to its transparency- with a complete word list publicly available- and its rule-based approach, which helps minimize inconsistency (Hutto & Gilbert, 2015). Figures 8, 9, 10 and 11 demonstrate these results.

Audit Methods

To assess and document biases within the dataset and pipeline, several diagnostic figures were produced. The composition of the data is addressed through article counts per event (Figure 3), regional distribution by outlet origin (Figure 4), and language breakdown by publisher (Figure 5). Model performance is evaluated via topic size distribution, outlier rates by event, and per-event topic counts (Figure 7). Sentiment is audited across publishers (Figure 8), across protest events and political leanings (Figure 9), by outlet ideological grouping (Figure 10), and by topic in the global model (Figure 11). Together, these figures allow the reader to identify

where the data is lacking, which publishers or regions dominate, and whether the sentiment classifier behaves consistently across ideological groups.

Figure 7

Topic Model Performance Diagnostics

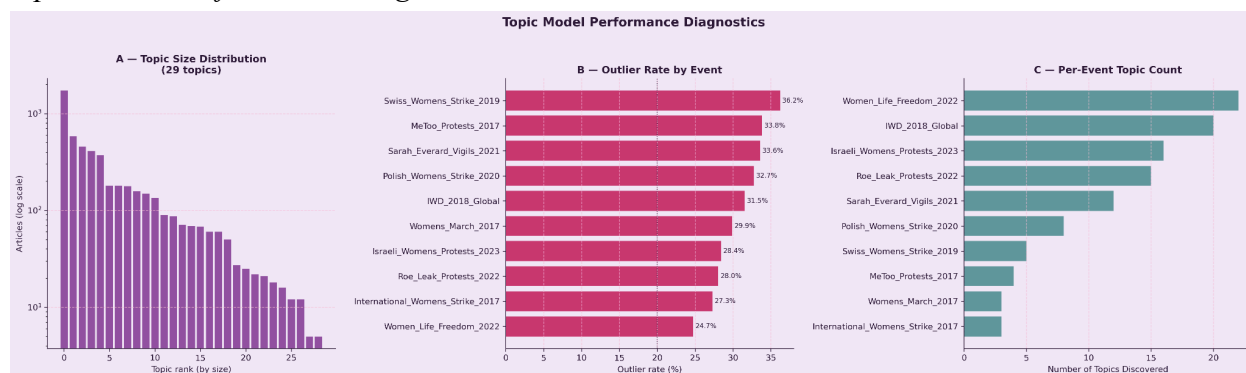


Figure 8

Mean Sentiment Score by Publisher

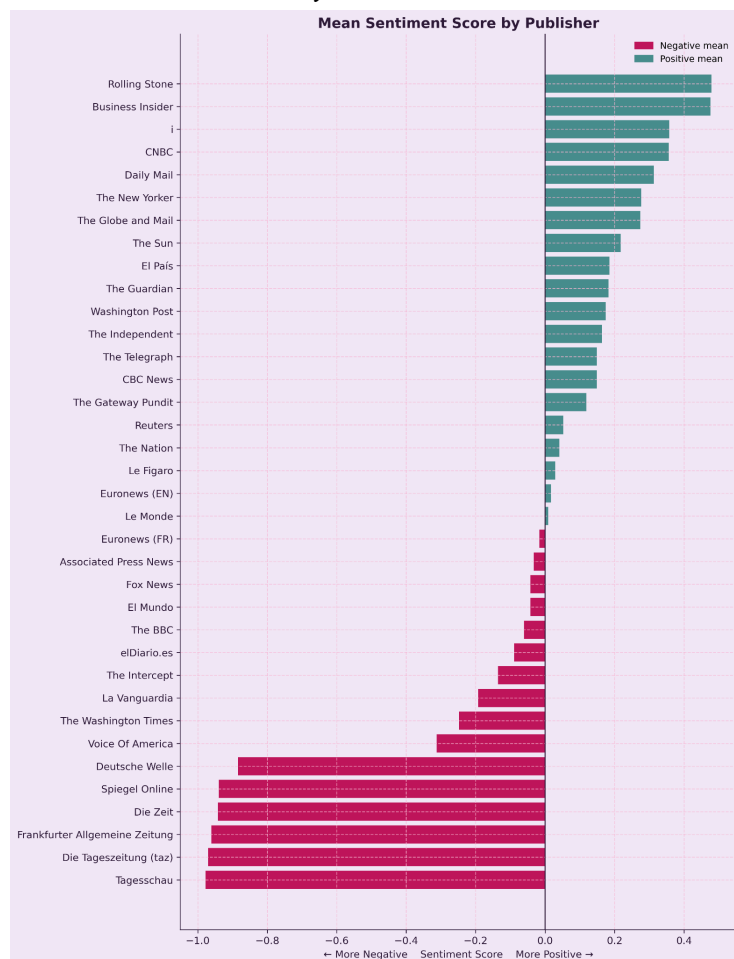


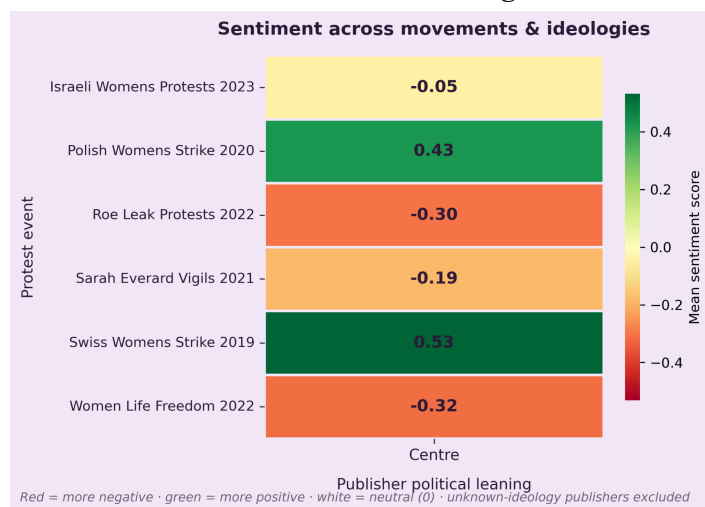
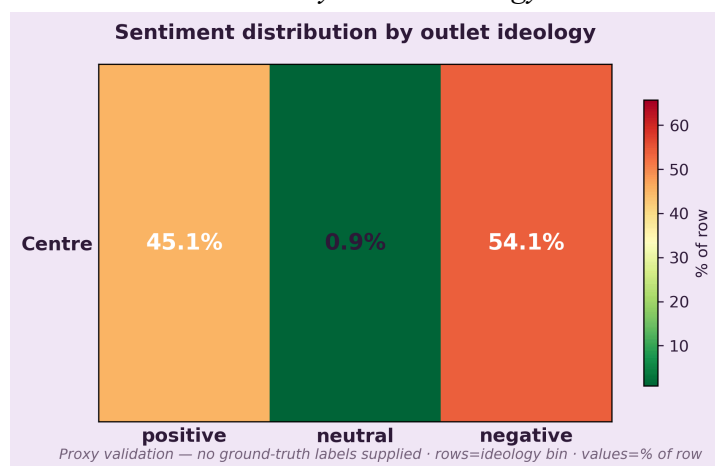
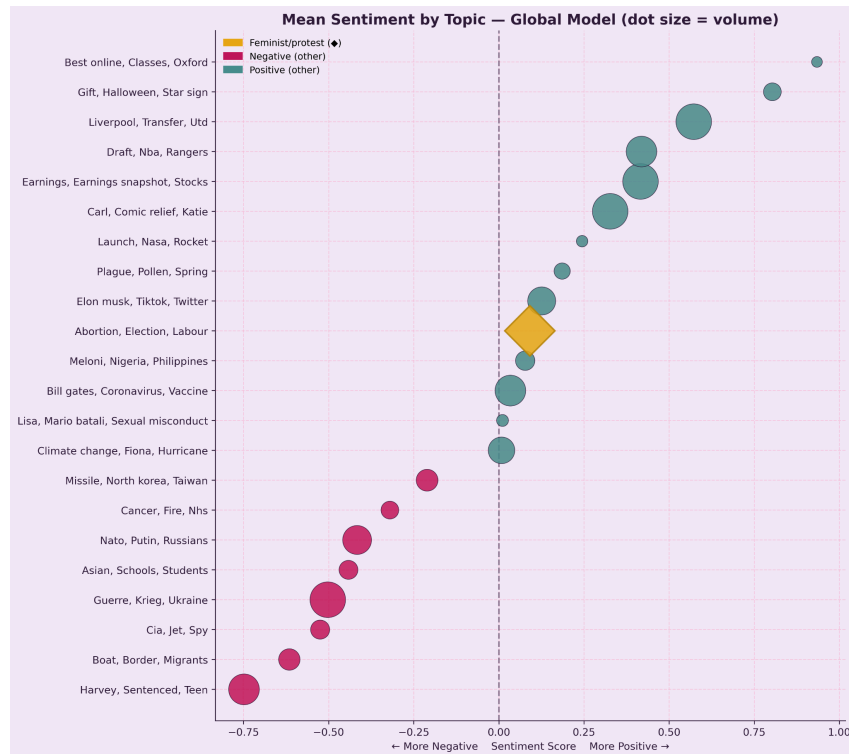
Figure 9*Sentiment Across Movements & Ideologies***Figure 10***Sentiment Distribution by Outlet Ideology*

Figure 11
Mean Sentiment by Topic - Global Model



Qualitative Methods

Desk Research

To contextualize the quantitative findings, desk research was conducted alongside the computational analysis. Sources included the Carnegie Global Protest Tracker for Gender and other prior research on feminist movements, protest media framing, and news representation (2026). This literature informed both event selection and the interpretation of the sentiment and topic patterns identified by the pipeline.

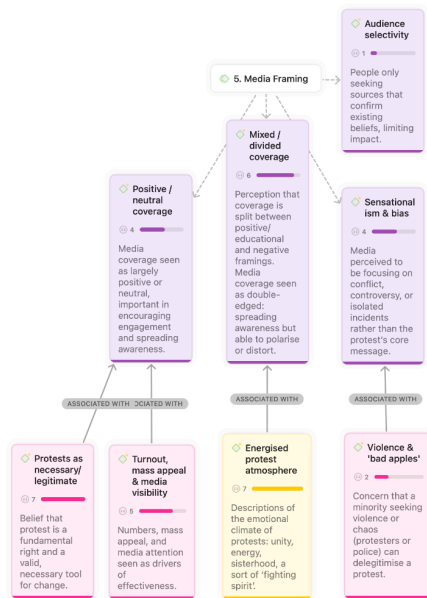
Interviews

The qualitative component was originally designed to include survey and ethnographic data collection at feminist events and community initiatives. However, due to limited access and non-response from several organisations, these approaches could not be implemented. As a result, the qualitative data was collected exclusively through seven semi-structured interviews.

Interviews were recorded, transcribed, and coded using Atlas.ti. The analysis focused on three core themes: perceptions of media connotation, thematic focus of coverage, and the role of media visibility in shaping engagement with feminist issues. The coding framework was informed by existing literature and developed iteratively during analysis (Braun & Clarke, 2006; Entman, 1993; Stroud, 2008). For the interview guide, see Appendix F.

In total, we conducted seven interviews, six with university students who engage regularly with news and feminist topics, and one with a university professor with expertise in feminist studies. Interviews were recorded, transcribed, and coded by hand using Atlas.ti, producing the map shown in Figure 13.

Figure 13
Qualitative Interview Coding Map (Atlas.TI)



Alternative Methods

A full failure analysis documenting alternative methodological approaches that were evaluated and either refined or entirely discarded – including body-text-based topic modelling, multilingual embedding without translation, and fixed versus adaptive minimum topic size– is provided in Appendix G.

Audit Results

Baseline Model Performance

To assess baseline model performance, we built a pipeline combining article collection, translation, sentiment analysis, and topic modelling that captured the entire news environment during each protest week and compared it with matched control weeks. This allowed us to examine how feminist protest events were situated within broader media and public discourse.

As shown in Figure 3, article collection was uneven across events. Women Life Freedom 2022, Israeli Women's Protests 2023, and Roe Leak Protests 2022 each had more than 1,000 articles,

while several earlier events had fewer than 150 articles. This imbalance affects the baseline model because larger event corpora are more likely to produce stable and diverse topic clusters.

At the topic modelling level, BERTopic was limited to about 30 topics across the corpus, as shown in Figure 7. Topic sizes were uneven, ranging from over 1,000 articles in the largest cluster to fewer than 10 in the smallest clusters. Outlier rates ranged from 24.7% to 36.2%, indicating that a substantial proportion of articles were not assigned to stable topic clusters before outlier reduction. In addition, the number of topics identified per event ranged from 2 to 22, suggesting considerable variation in thematic diversity across protest events.

Bias Detection Results

The audit focuses on representational inequality in feminist protest news coverage, examining which narratives are amplified and which are underrepresented in the dataset.

Five types of bias were identified. Representation bias arises from the selection of 45 publishers and article caps, limiting coverage of smaller outlets. Methodological bias results from preprocessing decisions such as translation, stopword removal, and topic modelling. Framing bias is introduced by the event-based sampling of predefined protest windows. Model bias relates to limitations in sentiment analysis and clustering in multilingual data. Finally, annotation bias may occur during qualitative coding.

Bias Reduction Techniques

Multiple methods were employed to mitigate the impact of bias in the dataset, methodology, and interpretation of results. These techniques did not remove bias completely, but they made the audit more transparent and helped us interpret the model outputs more carefully.

To reduce representation bias, we included publishers from different countries and different media contexts to increase the diversity of viewpoints represented by the dataset. We also set a uniform cap on the number of articles from all publishers to reduce the dominance of highly active news outlets (Mehrabi et al., 2019). In addition, control weeks were included to compare feminist protest periods with ordinary news periods. These steps made the sample more balanced.

To reduce methodological bias, we compared sentiment analysis results across publishers, language groups, and protest events to identify potential inconsistencies. Furthermore, we used BERTopic instead of an LLM model for topic modeling. This allowed us to inspect topic clusters through keywords, topic labels, and article examples.

To address annotation bias, interview data were coded and reviewed by at least two researchers to reduce individual subjectivity (Snow et al., 2008). Additionally, sentiment and topic analysis

results were compared across ideological groupings and showed consistent patterns across outlets (Figure 10), suggesting no evidence of systematic ideological bias (Mehrabi et al., 2019).

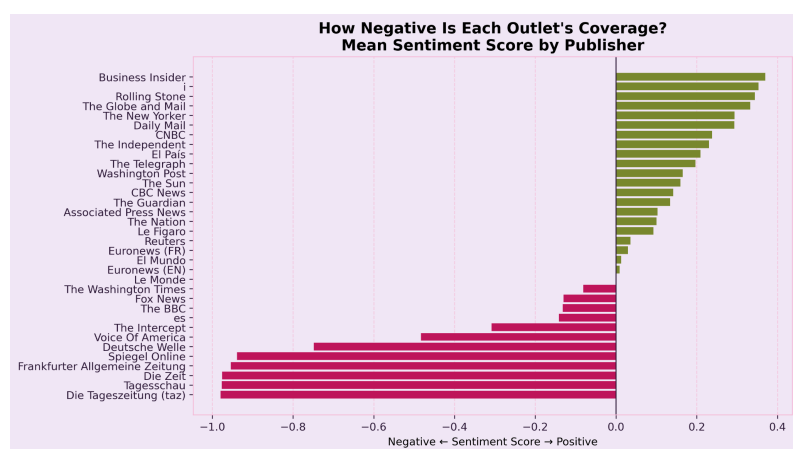
Framing bias was mitigated by the use of control weeks. By comparing protest-period articles to baseline coverage from the same publishers during non-protest periods, the study avoids the presumption that women's protest events receive uniquely negative coverage (Lopez Bernal et al., 2018).

Evidence of Disparities Across Groups

The most striking disparity was found across language groups. As shown in Figure 14, German-language outlets received sentiment scores close to the minimum of the scale, with Die Tageszeitung and Tagesschau close to -1.0. English-language outlets covering the same events scored near neutral or positive, such as the BBC -0.06 and Rolling Stone +0.44. Such large differences cannot be interpreted as direct evidence of language bias on their own, because outlet style, event selection, and article content may also influence sentiment scores (Galeazzi et al., 2024).

Figure 14

Mean Sentiment Score by Publisher



Second, topic modelling revealed disparities in the visibility of different types of news topics. As shown in Figure 6, the topic related to the war in Ukraine accounted for 22.6% of clustered articles. In contrast, feminist protest-related narratives did not appear as a clearly dominant topic in the full corpus. The closest related cluster was “Abortion, Election, Labour” but this topic was mixed with broader political and media narratives.

Fairness metric outcomes

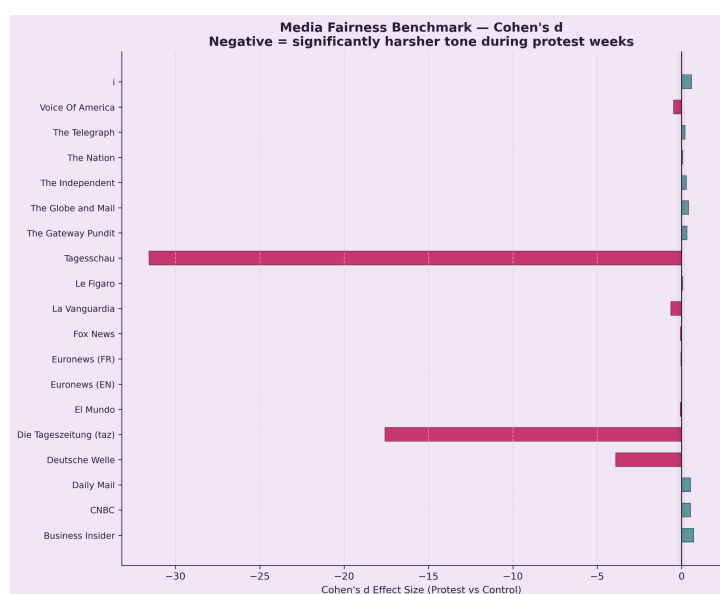
In this project, we used fairness metrics that were more suitable for evaluating sentiment patterns and representational visibility in media coverage. The fairness metric outcomes showed that

sentiment was not consistent across all groups. As shown in Figure 9, the Swiss Women's Strike 2019 scored about +0.53, while Women Life Freedom 2022 scored about -0.32. This suggests that different feminist protest events were represented with different sentiment tones.

Figure 15 shows the sentiment difference between protest weeks and control weeks. For most publishers, the difference was small. However, Tagesschau and Die Tageszeitung showed much more negative sentiment during the protest weeks. This result should be interpreted carefully, because it may also be influenced by sample size, language processing, or outlet style. Overall, these results show that sentiment outputs were not equally consistent across all events and publishers.

Figure 15

Media Fairness Benchmark



Impact of Bias Reduction and Comparative Visualizations

The comparative visualizations were used to connect the baseline limitations with the results after mitigation. Figure 3 shows that the number of collected articles was very uneven across protest events. This means that some larger events and more active publishers could have had a stronger influence on the results. After topic processing and outlier reduction, Figure 7 reveals that a substantial proportion of articles remained as unassigned outliers across all events. However, the topic sizes were still uneven, which means that broader mainstream news topics were still more visible than smaller feminist protest-related themes. A similar pattern appears in the fairness charts (Figure 9, Figure 15). Sentiment differences persisted across events and publishers, and between protest and control periods.

Overall, the mitigation strategies improved transparency and comparability but did not eliminate structural bias in the dataset. This reflects broader limitations of computational systems, which tend to privilege dominant narratives and reduce visibility of minority perspectives (Helm et al., 2024). Similarly, the limited size of the interview sample restricts the generalizability of the qualitative findings. Annotation uncertainty also remains inevitable in qualitative coding of politically sensitive data (Van Der Velden et al., 2025).

Contextualisation of Results

Linking Qualitative and Quantitative Insights

This audit combines computational text analysis with qualitative interpretation to contextualize the quantitative findings. Following critical data science approaches, quantitative modelling is understood as inherently interpretive, with choices in data selection, preprocessing, and model design shaping analytical outcomes (Eitzel, 2021). Because concepts such as fairness and meaning are contested, computational outputs require qualitative interpretation to be analytically meaningful (Madaio et al., 2020). Sentiment analysis in particular benefits from validation against human interpretation, as prior work has shown that automated measures can misrepresent contextual meaning (Cero et al., 2021).

In this study, qualitative interviews provide insight into how news audiences perceive feminist protest coverage. Participants generally did not view media reporting as neutral, but as shaping the perceived legitimacy and visibility of feminist causes. Most interviewees described a mix of awareness-raising and sensationalism in coverage, with protest visibility seen as closely linked to perceived impact and mobilisation.

These perceptions align with the quantitative findings. Feminist topics increased from 1.6% of topic distribution during control weeks to 12.9% during protest weeks, while broader global topics such as geopolitical conflict remained dominant across both periods. Sentiment scores also varied across events and linguistic contexts, reinforcing the importance of combining computational and qualitative perspectives when interpreting media representation.

Analysis of Structural Inequality and the Status Quo

Feminist critical theory provides an appropriate lens for interpreting these findings. Drawing on situated knowledge (Haraway, in Hancox-Li & Kumar, 2021), this perspective rejects the “view from nowhere”: journalists, editors, and the builders of sentiment tools are situated actors, not neutral transmitters of knowledge. Inequality can be reproduced through seemingly neutral routines. Journalistic practice creates hierarchies across protest topics preserving the status quo – refer to Appendix A (Brown & Harlow, 2019). Intersectionality (Crenshaw, in Wang et al., 2024; Winkle et al., 2023) is central: underrepresentation compounds for movements led by women of colour, non-English speakers, and the Global South. and Differential treatment becomes discrimination when systematic and persistent (Barocas et al., 2024). This supports H2: mainstream outlets need not be explicitly anti-feminist to marginalise protest; the sheer volume

dominance of normalised content (business, sports, institutional politics) dilutes coverage. Across every protest event examined, the largest share of articles in the corpus fell not under a women's-rights or protest topic but under unrelated categories – generic political news, the Ukraine war, sport, entertainment – while only one modelled topic corresponded substantively to feminist mobilisation. The protests were not absent from the news so much as structurally outnumbered, their disruptive potential absorbed into the categories that dominate the everyday news environment. Poell (2020) explains the underlying mechanism: the 24-hour news cycle rewards episodic, spectacle-driven coverage and structurally disfavours sustained attention to social causes. On the other hand, 5 interviewees explicitly tied protest impact to visibility and mass appeal to media coverage.

The framing of feminist protest cannot be read as neutral. News media frequently deploy delegitimizing frames to “uphold the status quo through the ‘protest paradigm’” (Mendelsohn et al., 2024), trivialising protesters by focusing on their methods rather than message (McLeod & Detenber, 1999). Recent research complicates this lens: delegitimization is unevenly distributed, with feminist protests treated as more legitimate than racial-justice ones (Brown & Harlow, 2019; Kilgo & Harlow, 2019). Gruber (2023) and Mourão et al. (2021) similarly find substantial sympathetic coverage alongside the delegitimizing pattern. This matters for H1: a blanket claim of negative sentiment is difficult to sustain. The interview and sentiment results reflect this and qualify H1 rather than confirm it. Sentiment varied by language, outlet, and event, with violence-centred protests such as the Sarah Everard vigils attracting markedly more negative coverage than broad, celebratory mobilisations like the Women's March. Strikingly, German-language outlets scored as overwhelmingly negative while English coverage was mildly positive – a divergence large enough to probably reflect the sentiment tool's cross-lingual calibration more so than any editorial difference in coverage between countries. It is worth noting too that positive sentiment does not necessarily indicate legitimisation. De Benedictis et al. (2019) found #MeToo covered positively in 56% of articles, yet this masked depoliticization – coverage that absorbed feminist messages into celebrity culture and individualist “empower women to speak out” narratives.

Interpretation of Computational Results and Sources of Bias

The computational results show two consistent patterns in feminist protest coverage. First, feminist protest narratives rarely emerge as distinct topics within the broader news ecosystem. Instead, they are either absorbed into mixed political categories or appear within large residual outlier clusters, while unrelated topics such as geopolitics and institutional news dominate most protest periods. This suggests that feminist protest coverage is structurally present but thematically diluted within mainstream news agendas, limiting its visibility as a coherent discourse (RQ1).

Second, sentiment analysis reveals substantial variation across linguistic and outlet contexts. While some feminist protest events are associated with neutral or moderately positive framing, others - particularly in certain language groups - receive more negative sentiment scores. German-language outlets in particular tend to show more negative sentiment compared to English-language coverage of similar events, indicating that framing differences are not consistent across media systems.

Taken together, these results indicate that feminist protest visibility is both uneven and highly context-dependent. Rather than being uniformly marginalized through negative framing, feminist movements appear to be differentially represented depending on language, outlet, and event type. At the same time, their limited thematic consolidation within topic models suggests that sustained visibility in mainstream news discourse remains constrained, with most attention absorbed by broader, more dominant news categories (RQ2).

Discussion

Benefits and Drawbacks

Media-framing scholarship has documented how coverage marginalizes movements, though less attention has been paid to critically evaluating the measurement tools used to study these patterns. Computational text-analysis research has similarly grown rapidly but faces ongoing concerns about whether its methods accurately capture the social constructs they aim to measure (Birkenmaier et al., 2024). A recurring conclusion in the critical data science literature is that automated methods benefit from human and theoretical interpretation alongside computational outputs (Grimmer et al., 2022). This study builds on this perspective by examining how feminist protest representation is shaped not only by media coverage itself, but also by the analytical tools used to study it. The framework combines multiple pipeline auditing steps with qualitative interview insights to provide a multi-layered interpretation of media representation (Blee & McDowell, 2022; Madaio et al., 2020). Despite implementing several mitigation strategies, the analytical pipeline remains affected by structural biases that cannot be fully eliminated. These limitations are inherent to the data, modelling choices, and framing assumptions underlying the study design.

Potential Impact on Social Inequality

Focusing on the domains of civil society and critical political movements, audit's empirical results provide clear evidence that media institutions reproduce structural social inequalities, thereby directly substantiating Hypothesis 2 (H2) and addressing Research Question 2 (RQ2). The thematic distribution in Figure 6 shows that during major feminist protest weeks, the dominant macro-theme across the wider news environment was the war in Ukraine, accounting for 22.6% of the corpus. Conversely, narratives substantively corresponding to core feminist political demands were systematically diluted, relegated to isolated micro-topics or embedded

within normalized, non-protest categories such as the mixed "Abortion, Election, Labour" cluster.

Mainstream media organizations do not need to adopt an explicitly anti-feminist editorial stance to marginalize civil society and critical political movements (Brown & Harlow, 2019). Instead, the structural dominance of standard journalistic routines (which heavily privilege high volumes of normalized coverage regarding business, sports, and institutional politics) is sufficient to dilute the disruptive potential of public mobilization (Poell, 2020). When examining these analytical text pipelines within the shifting paradigms of law and governance, the uncritical deployment of automated systems by policymakers introduces severe epistemic risks (Aysolmaz et al., 2023). Tool-level language biases that systematically misinterpret non-English narratives inevitably compound with density-based clustering thresholds that discard minoritarian topics. This algorithmic interaction renders the radical systemic demands of non-English-speaking communities invisible within digital discourse spaces (Hancox-Li & Kumar, 2021).

Opportunities for Improvement

This study utilized a sequential data pipeline that machine-translates non-English text prior to lexicon-based scoring. The stark cross-lingual disparities observed in this audit highlight the structural limitations of this traditional framework. Future research should instead deploy native multilingual sentence encoders or advanced multilingual large language models. Furthermore, following Noble's (2018) critique of algorithmic neutrality, the deflated sentiment scores of German outlets in Figure 14 require careful re-evaluation.

Similarly, the high volume of unassigned documents captured during text preprocessing highlights a clear need to reform how topic prevalence is modeled. The configuration mechanisms of unsupervised density-based clustering models require systematic re-evaluation. Future iterations must introduce dynamic or hierarchical topic-retention thresholds. Implementing these adaptive parameters is critical to preventing systematic statistical erasure. This structural safeguard ensures that localized, diverse activist themes representing vulnerable groups remain fully visible within digital discourse spaces.

Future Implications

The implications of this audit extend beyond feminist protest coverage to the broader trajectory of automated text analysis in the social sciences and protest research. As these pipelines become standard tools for tracking public discourse, measurement errors grow more serious. When sentiment scorers and clustering models are used at scale without scrutiny, they can quietly erase non-English and minority voices from the very policy tools meant to capture them (Ananny & Crawford, 2018). This is not a minor technical issue: research shows that translating text before scoring it can actually import and worsen bias rather than remove it (Goldfarb-Tarrant et al., 2023). Algorithms do not simply describe public debate; they shape whose voices are seen (Noble, 2018).

New rules such as the EU AI Act and GDPR make this urgent (Aysolmaz et al., 2023). Because bias builds up at every stage – from choosing sources to cleaning text to designing models – accountability means auditing each step, not just one part (Metzler et al., 2024). Future work should favour multilingual methods over translate-then-score designs and consider whether it is possible to keep small, local themes instead of discarding them. Pairing computational power with careful human interpretation (Eitzel, 2021; Madaio et al., 2020) is necessary to ensure tools do not disproportionately disadvantage marginalized publics.

Conclusion

This study produced two main findings. Feminist protest narratives rarely emerged as a distinct topic, instead collapsing into broader categories or appearing within outlier clusters, while unrelated news topics dominated most protest periods. In addition, sentiment scores varied substantially across linguistic contexts, with German-language outlets showing more negative scores and English-language outlets tending toward neutral or positive framing.

At the same time, the relative scarcity of feminist topics during non-protest periods raises further questions about the visibility of feminist issues in everyday news coverage and whether this limited baseline attention may constrain broader public engagement with the movement. Additionally, Figure 11 indicates that publishers tend to portray these events in a slightly more positive light during protest periods. This may suggest that feminist protests are generally framed as legitimate or accepted forms of civic participation rather than as disruptions to the status quo.

Through a range of auditing procedures and methodological safeguards designed to reduce bias and improve fairness, we have sought to represent the data as accurately as possible. Nevertheless, the study remains subject to several limitations, as discussed throughout the report. Future research could build upon this work by expanding the dataset, incorporating additional publishers or protest events, or focusing on more specific case studies in order to uncover patterns and impacts that may not be visible at a broader scale.

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Appendices

Appendix A

Appendix A: Project Outline Document

The project outline document was updated weekly throughout the project and therefore reflects its development over the course of the semester. Consequently, some information within the document may be outdated or inaccurate. The attachment of the original is intended to illustrate the project's development and growth over time. The contents for Weekly Goal 3 are presented here to further understanding of the status quo; for the complete original, please check the attachments.

The file can be found attached to the submission as “project_outline_document.pdf”.

Weekly Goal 3: Status Quo

The status quo theory, also known as status quo bias, holds that people prefer their current state of affairs and are somewhat resistant to change (“Status Quo Bias”, 2025). In liberal democracies, this is exemplified by political and social institutions that maintain existing power structures while formally allowing for a diluted form of public protest. Within these institutions, democratic processes such as elections, legislature, and court proceedings are considered the proper methods through which change should occur. While protests may be outwardly tolerated, they are considered an ‘outlier’ to these traditional democratic processes and thus are subjected to arbitrary constraints. Martin (1986) argues that liberal democracies manage this dissent through a combination of regulations and media framing that allow protests to coexist with democratic governance while limiting their capacity to disrupt the norm.

Another critical element of the status quo is that institutional entities wield greater influence than the bourgeoisie, thereby encouraging minority groups to adopt protest as an alternative political strategy. This can be seen in movements for gender equality. While liberal democracies promise fairness, systemic inequalities are maintained and recreated by long-standing institutions. Researchers argue that these very inequalities – and the government’s failure to address them – serve as the spark behind many feminist movements. For example, Portillo and Bono (2021) illustrate how feminist activism in Spain has transformed from a desire to be more politically involved to a raw, massive movement of bodies. This large-scale physical demonstration was fueled by a seeming insufficiency in formal political processes to adequately address patriarchal structures deeply embedded within the government.

While these empty promises of equality can be the initial catalyst in spurring protests, if institutions continue not to address these issues, the desire for protest is magnified and persists. For example, the 1991 Swiss Women's Strike. This strike was organized in response to frustration that the constitution's guarantees of gender equality, adopted in 1981, had not been translated into tangible improvements in wages or workplace rights (Abplanalp, 2019). Similarly, feminist strikes across different countries, including Spain and Poland, have also been partially motivated by slow action (or lack thereof) to address gender inequalities (Campillo, 2019; Król & Pustulka, 2018; Kubisa & Rakowska, 2019).

In short, protests are often the byproduct of a failure to deliver meaningful change on the part of the government. While liberal democracies allow for protest, they also determine how forms of protest are manifested, perceived, and managed. Hence, ordinary people have a semi-regulated method through which they can attempt to achieve tangible change. This project's analysis demonstrates that mainstream media networks marginalize critical political movements not through overt editorial opposition, but through a structural dominance of normalized news content that systematically dilutes the disruptive potential of public mobilization (Brown & Harlow, 2019; Poell, 2020; see full context in the Analysis of Structural Inequality and the Status Quo section). Furthermore, the study qualifies traditional assumptions of the protest paradigm. Instead of reflecting a blanket pattern of negative delegitimization, the empirical results reveal that media sentiment configurations are highly variable across distinct linguistic and event-specific contexts (De Benedictis et al., 2019; Mendelsohn et al., 2024).

Appendix B

Annotated Data Source List

Common Crawl Foundation. CC-News.

CC-News is an open-access archive of online news articles collected through Common Crawl. This was used as the project's primary dataset for collecting multilingual news coverage surrounding women's protest events and matched control periods. It was chosen due to its international scope, transparency, and compatibility with computational text analysis. Limitations include uneven coverage- particularly in earlier years- and overrepresentation of larger news organizations. Further details regarding its usage and implementation can be found in the Data and Methods section above.

Appendix C

Stakeholder Analysis

This stakeholder analysis identifies the main groups affected by the algorithmic analysis of feminist protest coverage.

Stakeholder	Project role	Relevance to social inequality
Feminist activists	They are the groups being represented in news coverage.	Unequal visibility can weaken feminist movements and reduce public recognition. If feminist protests are underrepresented, their political demands may become less visible.
News audiences	They receive information about feminist protests through the news media.	Public understanding of gender equality issues depends on how protests are represented in the media.
News publishers	They produce the news articles included in the dataset.	Media institutions can reproduce structural inequalities through selective attention and framing.
Research team	They design the methods, choose the data and interpret results.	Methodological choices can shape how inequality is measured and explained. Publisher selection, translation, topic modelling and sentiment analysis can affect the results.
Policy makers	They may use research on algorithmic systems, media data and social inequality.	Regulation can address how systems reproduce unequal representation and visibility.
Other marginalised groups	Other social movements may also be affected by similar algorithmic and media bias.	The findings show that algorithmic visibility is not only a technical issue, but also a structural inequality issue.
CC-News Data providers	They provide the dataset used for computational analysis.	Dataset structure can make some regions, languages and movements more visible than others.

Appendix D

Ethics Assessment Results and Ethics Documentation

This project involved both computational analysis of news data and social science research with participants. The computational part used publicly available news data from CC-News. The main ethical risks in this part were related to representation, language bias, and the possibility of reproducing existing inequalities in news coverage. The social science part involved surveys and interviews. The main ethical risks were informed consent, privacy, anonymisation, and responsible data storage. Participants were informed about the purpose of the project before taking part. Participation was voluntary, and participants could refuse to answer questions or withdraw. Personal identifiers were removed from the analysis and findings were reported in an anonymised way.

The full consent forms and research materials are included in Appendices E, F and G.

Ethical issue	How we addressed it
Informed consent	Participants received information about the project before taking part. Consent was collected before the survey and interview.
Privacy	Personal identifiers were removed from the analysis. The report does not identify individual participants.
Data storage	Participant data was only used for this course project and was accessible only to the research team.
Participant harm	The questions focused on media representation and feminist protest coverage. Participants could skip questions or withdraw.
Representation bias	The project recognised that the news dataset was not equally balanced across protests, regions, languages and publishers.
Language bias	Non-English news content was translated into English before analysis. This may affect sentiment analysis and topic modelling results.
Methodological bias	The research team discussed limitations from publisher selection, event selection, translation, sentiment analysis and topic modelling.
Transparency	The report explains the data-processing pipeline, audit methods and limitations of the computational analysis.

Appendix E

Survey Questionnaire and Consent Form

Representation and Diversity in Web Content- This study is being conducted by student researchers in Computational Social Science at the University of Amsterdam (UvA).

Goal of the study- The aim of the study is to look into the distribution and representativeness of news coverage on feminist protests. The information is needed to create a final research report that pairs qualitative and quantitative research methods.

Procedure of the study- If you agree to participate, you will be asked to take part in a survey. You will be asked several questions – either open-ended or multiple choice – to gauge your personal experience with and opinion on a few topics. We ask questions about feminism, feminist protests, and the media coverage of these events.

The survey should take approximately 5 - 10 minutes.

Voluntary participation- Your participation in this study is voluntary: you are not obligated to participate. You may decide to stop your participation at any time during the study, and you may also decide to withdraw your data after the study. If you decide to stop or withdraw, your data will no longer be used for the research and will be deleted. You do not have to provide a reason for this, and it has no disadvantages for you.

Privacy- Your responses will be kept confidential. All data will be securely stored and only accessible to the research team and directly relevant UvA instructors. Any identifying information (such as names or specific details) will be anonymized in our analysis and in any published reports or presentations.

We use some third-party software to store and process your data. UvA has contracts with these parties to guarantee your privacy:

- Your data will be securely stored with Microsoft OneDrive and/or Canvas.
- Your data will be processed by Microsoft Forms to store data, analyze data, collaborate, manage questionnaires, etc.

Additional third-party software that we use:

- Atlas.ti for coding survey responses.

Data retention period- Personal data is kept in a closed and secure archive for as long as necessary for research and educational purposes, but for a maximum of 10 years (from the moment of collection).

Further information- If you have any questions about the study, either before you participate or afterwards, please feel free to contact the responsible researcher: Greta Pfeiffer, greta.pfeiffer@student.uva.nl.

- If you have any formal complaints about this research, please send an email to the UvA staff member supervising this project: Kunhe Li, k.li2@uva.nl. The staff member that approved the study will consider your complaint and contact you as soon as possible.

DECLARATION OF CONSENT- With this, you declare:

- I am 16 years or older.
- I have read and understood the information.
- I agree to participate in the study and to use the data obtained with it.
- I reserve the right to withdraw this consent without giving any reason.
- I reserve the right to stop the study at any time I wish.

1. To participate in the survey, click on “Yes” below.

- Yes, No

2. What is feminism to you?

3. Would you consider yourself a feminist? Why or why not?

4. Do you have any prior experience with feminist protests? If not, with protests in general?

- Yes, with feminist protests
- Yes, with other protests
- No, with neither

5. Which protests have you attended? Please be as specific as possible.

6. How regularly do you attend these events?

- I attend... very often, often, sometimes, rarely, never

7. How involved would you consider yourself during these events?

- I am... extremely involved, very involved, moderately involved, slightly involved, very slightly involved

8. What are your thoughts on protests- are they good? Bad? Effective or ineffective?
9. Have you ever heard about feminism in the news? Feminist protests?
 - Yes, No
10. How often do you hear about it?
 - I hear about it... very often, often, sometimes, rarely, very rarely
11. Did you have any general thoughts or impressions on this coverage?
12. How do you think the topics of feminism/feminist protests are connotated/framed (ie, positively, negatively, etc) in the media? How about compared to other events?
13. Do you feel as though this coverage affected the cause in any way? (ie, by furthering or hindering it)
14. How trustworthy do you find news coverage about feminist protests?
 - Very trustworthy, somewhat trustworthy, neutral, somewhat untrustworthy, very untrustworthy
15. Do you have any other thoughts to share on feminism, feminist protests, or the coverage of these events?

Appendix F

Interview Guide

<i>Introduction:</i> • Goal: Understanding the interviewee's experience with feminism and feminist protests, gauging the general conditions surrounding these events, motivating the interviewee to share how they feel news and news coverage relate to these activities. • Intro: ○ "Hello! We are students at UVA in the field of Computational Social Science doing a study on the role of news content in feminist inequality and protests. We're hoping to assess your thoughts about these topics. Today we're going to discuss your past experiences, general thoughts on feminism, prior protesting experience, and opinion on the role of news in these contexts." • Briefly go through the key points of the Consent Form • Ask for a recording of the interview ("Is it OK for you if I record the interview?")	
<i>Subtheme 1: Background Info</i> <i>Goal:</i> Warm-up interviewee, gauge past experiences and knowledge within this field	Q1: Everyone has their own definition of feminism- what is it for you? Q2: Would you consider yourself a feminist? Why? Q3: (How) do you speak to people in your life about feminism? E.g., family and friends?
<i>Subtheme 2: Feminist Protests</i> <i>Goal:</i> Understand the interviewee's experience with protests and determine their impressions of them	Q4: Do you have any prior experience with feminist protests? If not, with protests in general? (make sure they answer with specific protests if they have) - Q4a: How regularly do you attend/how involved would you consider yourself? - Q4b: Can you give a brief description of the atmosphere at these events? - Q4c: What are your thoughts on protests- are they good? Bad? Effective or ineffective? - Q4d: Is there a specific factor or factors that determine how effective a protest is? - Q4e: Would you say that there are good and bad protests?
<i>Subtheme 3: News Coverage</i> <i>Goal:</i> Gain insight into the relationship between news/media and protests.	Q5: Have you previously read about feminism in the news? Feminist protests? - Q5a: Did you have any general thoughts or impressions on this coverage?

	- Q5b: How do you think the topics of feminism/feminist protests are connotated/framed (ie, positively, negatively, etc) in the media? How about compared to other events? - Q5c: Do you feel as though this coverage affected the cause in any way? (ie, by furthering or hindering it)
<i>Subtheme 4: Conclusion</i> <i>Goal:</i> Wrap up with a broad, thought-provoking question	Q6: What do you think is the future of feminism?
If time allows, ask the interviewee if they have any other thoughts or comments they would like to make. Wrapping up the interview. Thank the interviewee for taking the time to sit down with you.	

Appendix G

Interview Consent Form

Representation and Diversity in Web Content

This study is being conducted by student researchers in Computational Social Science at the University of Amsterdam (UvA).

Goal of the study

The aim of the study is to look into the distribution and representativeness of news coverage on feminist protests. The information is needed to create a final research report that pairs qualitative and quantitative research methods.

Procedure of the study

If you agree to participate, you will be asked to take part in an interview. You will be asked several questions to gauge your personal experience with and opinion on a few topics. We ask questions about feminism, feminist protests, and the media coverage of these events.

The interview should take approximately 20-30 minutes.

Voluntary participation

Your participation in this study is voluntary: you are not obligated to participate. You may decide to stop your participation at any time during the study, and you may also decide to withdraw your data after the study. If you decide to stop or withdraw, your data will no longer be used for the research and will be deleted. You do not have to provide a reason for this, and it has no disadvantages for you.

Privacy

Your responses will be kept confidential. All data will be securely stored and only accessible to the research team and directly relevant UvA instructors. Any identifying information (such as names or specific details) will be anonymized in our analysis and in any published reports or presentations.

We use some third-party software to store and process your data. UvA has contracts with these parties to guarantee your privacy:

- Your data will be securely stored with Microsoft OneDrive and/or Canvas.

Additional third-party software that we use:

- Atlas.ti for coding Interview transcriptions

Data retention period

Personal data is kept in a closed and secure archive for as long as necessary for research and educational purposes, but for a maximum of 10 years (from the moment of collection).

Further information

If you have any questions about the study, either before you participate or afterwards, please feel free to contact the responsible researcher: Greta Pfeiffer, greta.pfeiffer@student.uva.nl.

If you have any formal complaints about this research, please send an email to the UvA staff member supervising this project: Kunhe Li, k.li2@uva.nl. The staff member that approved the study will consider your complaint and contact you as soon as possible.

DECLARATION OF CONSENT

With this, you declare:

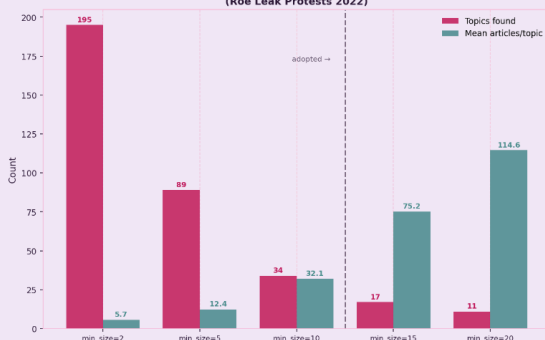
- I am 16 years or older.
- I have read and understood the information.
- I agree to participate in the study and to use the data obtained with it.
- I reserve the right to withdraw this consent without giving any reason.
- I reserve the right to stop the study at any time I wish.

A — Topic label quality: full body text vs title-only (Roe Leak Protests 2022)

Body text topics		Title-only topics (adopted)	
	Label		Label
Topic 0	de, die, der	Topic 0	Fur, Putin, Ukraine
Topic 1	de, que, la	Topic 1	★ Millones euros, Mujeres, Vestido
Topic 2	the, to, of	Topic 2	Charles, Cost living, Speech
—	(only 3 topics total)	Topic 3	Nato, Saudi, Sports
		Topic 4	Contest, Final, Song
		Topic 5	Daily, Summer, Teen mom
		Topic 6	Nine, Polis, Retour
		Topic 7	Bill gates, Covid, Taiwan
		Topic 8	League, Liverpool, Test
		Topic 9	★ Democrats, Senate, Supreme court
		Topic 10	Appeal, Killed, Mystery
		Topic 11	California, Climate, Debate

Body text topics dominated by wire-service boilerplate; protest-relevant topics buried at ranks 6 and 11. Title-only topics surface the core protest narrative at rank 1–2; feminist topics rescued by keyword detection (★)

B — Effect of min_topic_size
(Boe Leak Protests 2022)



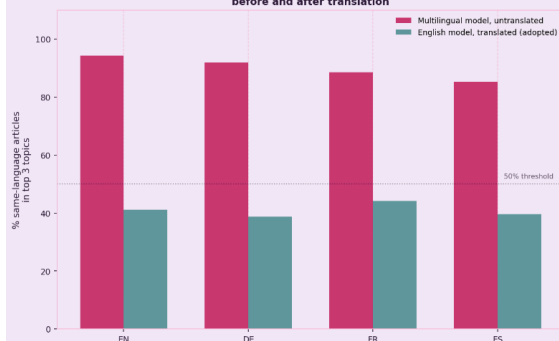
min. size=2 → 195 micro-clusters (avg 5.7 articles), min. size=15 (adopted) → 17 coherent topics (avg 75 articles)

C — Per-event model: fixed vs adaptive min topic size

Event	N	Fixed=15	Adaptive	Recovered
Womens March 2017	76	FAILED	min=2	4
Indi Womens Strike 2017	44	FAILED	min=2	3
MeToo Protests 2017	91	FAILED	min=3	6
WD 2018 Global	83	FAILED	min=3	5
Swiss Womens Strike 2019	312	8 topics	min=15	8
Polish Womens Strike 2020	899	8 topics	min=15	8
Sarah Everard Vigils 2021	1011	9 topics	min=15	9
Roe Leak Protests 2022	1426	17 topics	min=15	17
Women Life Freedom 2022	1540	23 topics	min=15	23
Israeli Womens Protests 2023	1568	19 topics	min=15	19

Four pre-2019 events failed under fixed `min_size=15` (sparse CC-News).
Adaptive retry recovered all events; fallback size flagged in figures.

D — Language cluster homogeneity: before and after translation



Multilingual embeddings produced language-homogeneous clusters (>85%)
Translating first reduced same-language co-clustering to ~40%.

E — Pipeline ordering: topic modelling before vs after translation

Original order	Adopted order
<code>topic_model ~ sentiment_analysis</code>	<code>sentiment_analysis ~ topic_model</code>
Embedding model: multilingual- <code>es-large</code> (1.1 GB)	Embedding model: all-MiniLM-L6-v2 (80 MB)
Input text: Raw multilingual titles	Input text: translated title (English)
Cluster quality: Language-homogeneous (Panel D)	Cluster quality: Thematically coherent cross-language
Top label (Role): Said, Women, According	Top label (Role): Abortion, Access, Pro choice
Feminist topics: 2 of 10 events flagged	Feminist topics: 7 of 10 events flagged
Runtime: ~45 min	Runtime: ~12 min

Reordering eliminated language-homogeneous clustering, reduced model size 93%, halved runtime, and increased feminist topic detection from 2 to 7 events.

Appendix I

Relevant Laws & Regulations

Law	Description
Directive on Copyright in the Digital Single Market (CDSM Directive)	<i>Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on copyright and related rights in the Digital Single Market</i> (European Parliament & Council of the European Union, 2019). Copyright normally gives authors exclusive control over the copying of their work. Because text and data mining (TDM) – which this project applies to the CC-News corpus – necessarily involves copying that text, it would ordinarily require permission from every rightsholder. The CDSM Directive resolves this by creating two exceptions that permit TDM without such permission, under different conditions. Article 3: the scientific research exception (applies to this project) Article 3 allows research institutions to carry out TDM for scientific research, provided they have lawful access to the material. Its defining feature is that it cannot be waived; rightsholders have no power to opt out. In other words, a university researcher with lawful access to news articles may mine them for scientific research, and a publisher cannot prevent this by attaching a “no mining” clause. As Margoni and Kretschmer (2022) explain, Article 3 is “imperative and finds application in cases of text and data mining for the purpose of scientific research by research and cultural institutions”, whereas Article 4 merely “permits text and data mining by anyone but with rights-holders able to ‘contract-out’”. Article 4: the general purpose exception (applies to commercial use) Article 4 extends a broader TDM permission to anyone, including

		commercial actors, again requiring lawful access. Unlike Article 3, however, rightsholders may opt out by expressly reserving their rights “in an appropriate manner, such as machine-readable means” (European Parliament & Council of the European Union, 2019, Art. 4(3)) – for example, through metadata or robots.txt-style signals that a crawler can detect. Where rights have been reserved in this way, a commercial miner must obtain a licence. <i>Relevance to the project:</i> This study is non-commercial scientific research conducted in a research setting, so it falls under Article 3, the non-waivable exception. Any “no mining” reservation a news outlet may have placed on its content does not apply. The Article 4 opt-out becomes relevant only if the model developed here were later deployed for a commercial purpose.
Artificial Intelligence (EU AI Act)	Act	<i>Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence</i> (European Parliament & Council of the European Union, 2024). The AI Act is the first comprehensive legal framework for AI and adopts a risk-based structure, sorting systems into four tiers: unacceptable (prohibited), high-risk, limited-risk (transparency obligations), and minimal-risk (European Commission, 2024). It entered into force on 1 August 2024 and applies in phases: prohibitions and AI-literacy obligations from 2 February 2025, general-purpose AI model obligations from 2 August 2025, and the bulk of remaining provisions from 2 August 2026 (European Commission, 2024). Subsequent amendments have postponed certain high-risk deadlines and adjusted the timeline for some transparency provisions (European Commission, 2024).

	<i>Relevance to the project:</i> The sentiment-analysis and topic-modelling pipeline used here is best characterised as a minimal-risk research tool: it analyses publicly reported news content and does not make consequential decisions about individuals, so it attracts no mandatory compliance obligations, though voluntary codes of conduct may be adopted. The transparency provisions become pertinent only if outputs incorporating generated or summarised text were to be published for an audience, in which case disclosure obligations would apply.
General Data Protection Regulation (GDPR)	<i>Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data</i> (European Parliament & Council of the European Union, 2016). The GDPR governs the lawful, fair, and transparent processing of personal data (Art. 5). It requires a lawful basis for processing (Art. 6), most relevantly informed consent (Art. 7). It provides a specific safeguard regime for processing undertaken for scientific-research purposes (Art. 89) stipulating appropriate technical and organisational measures – such as pseudonymisation – be in place. Personal data that has been rendered fully anonymous falls outside the Regulation’s scope (Recital 26). <i>Relevance to the project:</i> The computational phase draws on publicly available news data, which may nonetheless contain personal data (e.g., named individuals in reporting), so the principles of data minimisation and purpose limitation still apply. The GDPR is most directly engaged by the social-science phase: interview and survey data constitute personal data and require a clear lawful basis, informed consent, and responsible handling. The project addresses this through not collecting

	data through which participants could be identified, explicit consent procedures, and secure storage, consistent with the Art. 89 research safeguards. Notably, the AI Act does not displace the GDPR; both apply concurrently where AI systems process personal data (European Commission, 2024).
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Source: Own presentation.

Appendix J

Statement of Project Work Division

Team members: Chris Xia, Greta Pfeiffer, Laura Ponoran, GengXin Geng, Cristina Gomero

Date: February 17, 2026

TASK and DESCRIPTION	STATEMENT OF WORK	
Week Goals	<i>Short description of how the work was done, with names.</i>	
Week Goal 1: Week Goal 1: Data Exploration	Standup presentation was done by Greta Pfeiffer and Laura Ponoran. The presentation was created by Laura Ponoran, who worked on the first half, and GengXin Geng, who worked on the second half. Laura and GengXin updated the project outline document together.	
Week Goal 2: Research Question and Data Needs	Standup presentation was done by Cristina Gomero and Chris Xia. The whole group worked together to create a Data Science Flow Diagram. Greta Pfeiffer found external data sources and updated the project outline document. Greta and Cris worked together on the presentation.	
Week Goal 3: The Literature	Standup presentation was done by GengXin Geng and Laura Ponoran. Laura wrote the weekly goal, and GengXin worked on the slides and updated the project outline document.	
Week Goal 4: Method Exploration	Standup presentation was done by Greta Pfeiffer and Cristina Gomero. Laura Ponoran updated the project outline document with the help of the whole team. Cris worked on the slides.	
Week Goal 5: Synthesizing Status Quo Theory and Your Computational Model	Standup presentation was done by Chris Xia and GengXin Geng. GengXin worked on the slides, They also wrote the weekly goal.	
Week Goal 6: Critical Research Question and Social Science Research	Standup presentation was done by Laura Ponoran and Greta Pfeiffer. The project outline document was updated primarily by Laura Ponoran. The presentation was created by Greta – who also helped with designing the interview questions and reaching out to research participants/sites.	
Week Goal 7: Critical Theories	Standup presentation was done by Cristina Gomero and Chris Xia. They also wrote the weekly goal.	
Week Goal 8: Systematic Bias Audit	Standup presentation and weekly goal was done by GengXin Geng and Chris Xia. Meanwhile, Greta and Cristina coded interviews and Laura finished refining the quantitative data analysis algorithm.	
TASK and DESCRIPTION	LEAD CONTRIBUTOR	2 nd CONTRIBUTOR
Admin and coordination tasks		
Admin: Overview and Organization	Greta Pfeiffer	Cristina Gomero
Admin: Deadline and Submission Management	Laura Ponoran	GengXin

Admin: Event Planning and Coordination	Chris Xia	
<i>Code: please list the major code components or functional blocks in your project. There is space for four components, you may list more (or fewer) if needed for your project.</i>		
Component 0: Data cleaning and management, and null handling. Please give a brief description of what was achieved in this section: ingestion, storage, and structure CC-News articles across protest and control weeks	Laura Ponoran	
Component 1: Data collection and corpus construction Please give a brief description of what was achieved in this section: Event-based crawling of CC-News using Fundus, collecting articles across 10 protest events and matched control weeks	GengXin Geng	
Component 2: Language processing and sentiment analysis Please give a brief description of what was achieved in this section: Detects article language using Lingua, translates non-English articles into English (Helsinki-NLP opus-mt), and applies VADER sentiment scoring	Laura Ponoran	
Component 3: Topic modelling and semantic clustering Please give a brief description of what was achieved in this section: Applies BERTopic to translated English text to extract semantic topics at both global and per-event levels	Laura Ponoran	
Component 4: Bias analysis, diagnostics, and model validation Please give a brief description of what was achieved in this section: Implements systematic bias auditing across representation, framing, methodological, and model bias dimensions	Laura Ponoran	
Component 5: Visualisation and figure generation Please give a brief description of what was achieved in this section: Produces figures and summary outputs	Laura Ponoran	
Project Report		
Executive Summary	Chris Xia	
Introduction and Problem Statement	Chris Xia	

Background and Policy Relevance and subsequent components- Component 0: Algorithmic Bias and Social Inequality Component 1: Policy and Governance Relevance Component 2: Stakeholders and Structural Inequality Component 3: Theoretical Framework Component 4: Traditional Social Science Theory Component 5: Critical Theory Perspective	Cristina Gomero	
Data and Methods and subsequent components- Component 0: Research Design Component 1: Dataset Component 2: Data Collection Component 3: Publisher Selection Component 4: Quantitative Methods Component 5: Qualitative Methods Component 6: Alternative Methods	Laura Ponoran	
Audit Results and subsequent components - Component 0: Baseline Model Performance Component 1: Bias Detection Results Component 2: Evidence of Disparities Across Groups Component 3: Impact of Bias Reduction Component 4: Comparative Visualizations	GengXin Geng	
Contextualization and subsequent components - Component 0: Qualitative and Quantitative Methods Component 1: Structural Analysis of Inequality and the Status Quo Component 2: The Feminist Critical Theory Perspective Component 3: Interpretation of Computational Results and Sources of Bias Component 4: Interpretation of Computational Results Component 5: Sources of Bias	Greta Pfeiffer	
Discussion and Conclusion and subsequent components - Component 0: Benefits and Drawbacks Component 1: Potential Impact on Social Inequality Component 2: Opportunities for Improvement Component 3: Future Implications Component 4: Conclusion	Chris Xia	
Bibliography	Cristina Gomero	Greta Pfeiffer

Appendices	Laura Ponoran	Greta Pfeiffer
Project Outline Document	All	
Annotated Data Source List	Laura Ponoran	
Stakeholder analysis, updated.	GengXin Geng	
Ethics assessment results and any required ethics documentation.	GengXin Geng	
Survey Questionnaire and Consent Form	Laura Ponoran	All
Interview Guide	Laura Ponoran	Greta Pfeiffer
Interview Consent Form	Greta Pfeiffer	Cristina Gomero
Failure Analysis	Laura Ponoran	
Relevant Laws & Regulations	Greta Pfeiffer	
Statement of Project Work Division	Laura Ponoran	All